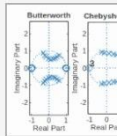
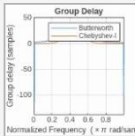
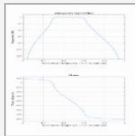
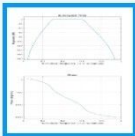
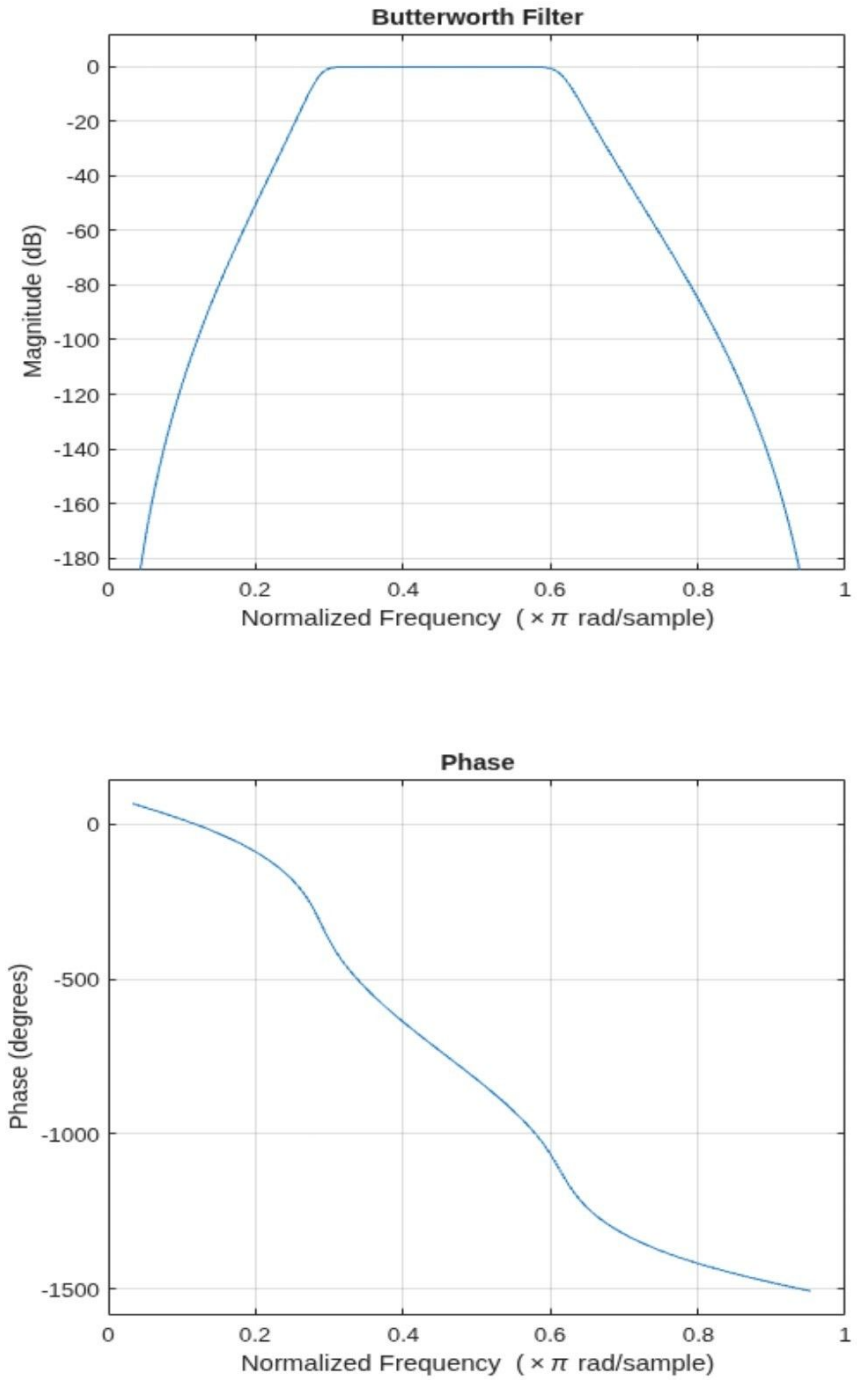




MATLAB Drive > Dsp3_1.m

```
1 clc;
2 clear;
3 close all;
4
5 % Specifications
6 Fs = 2000;
7 Rp = 1;
8 Rs = 40;
9
10 Wp = [300 600]/(Fs/2);
11 Ws = [200 700]/(Fs/2);
12
13 % Butterworth Filter
14 [n1,W1] = buttord(Wp,Ws,Rp,Rs);
15 [b1,a1] = butter(n1,W1,'bandpass');
16
17 % Chebyshev Type-I Filter
18 [n2,W2] = cheblord(Wp,Ws,Rp,Rs);
19 [b2,a2] = cheby1(n2,Rp,W2,'bandpass');
20
21 % Magnitude Response
22 figure;
23 freqz(b1,a1);
24 title('Butterworth Filter');
25
26 figure;
27 freqz(b2,a2);
28 title('Chebyshev Type-I Filter');
29
30 % Group Delay
31 figure;
32 grpdelay(b1,a1);
33 hold on;
34 grpdelay(b2,a2);
35 legend('Butterworth','Chebyshev-I');
36
37 % Pole-Zero Plot
38 figure;
39 subplot(1,2,1);
40 zplane(b1,a1);
41 title('Butterworth');
42
43 subplot(1,2,2);
44 zplane(b2,a2);
45 title('Chebyshev-I');
46
47 % Impulse Response
```

Figure 1: Dsp3_1





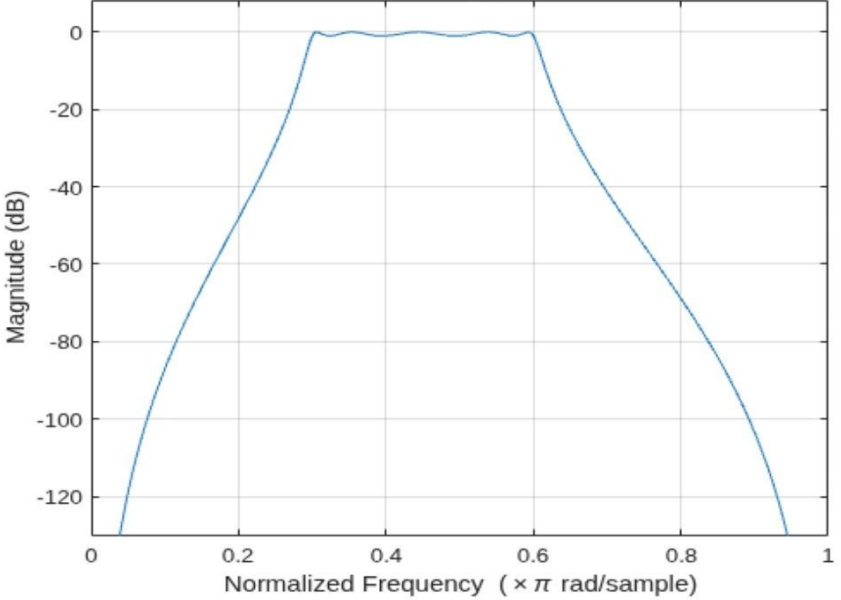
Figures

SHOW DATA CURSOR



Figure 2: Dsp3_1

Chebyshev Type-I Filter



Phase

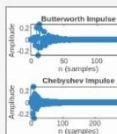
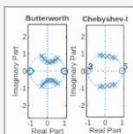
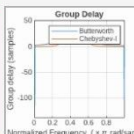
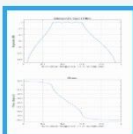
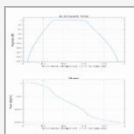
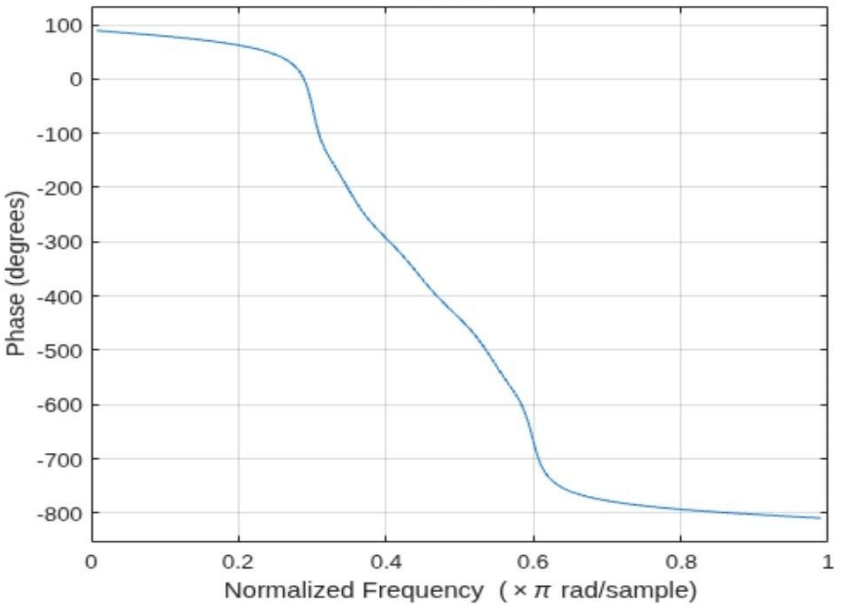


Figure 3: Dsp3_1

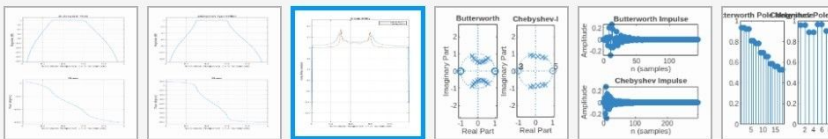
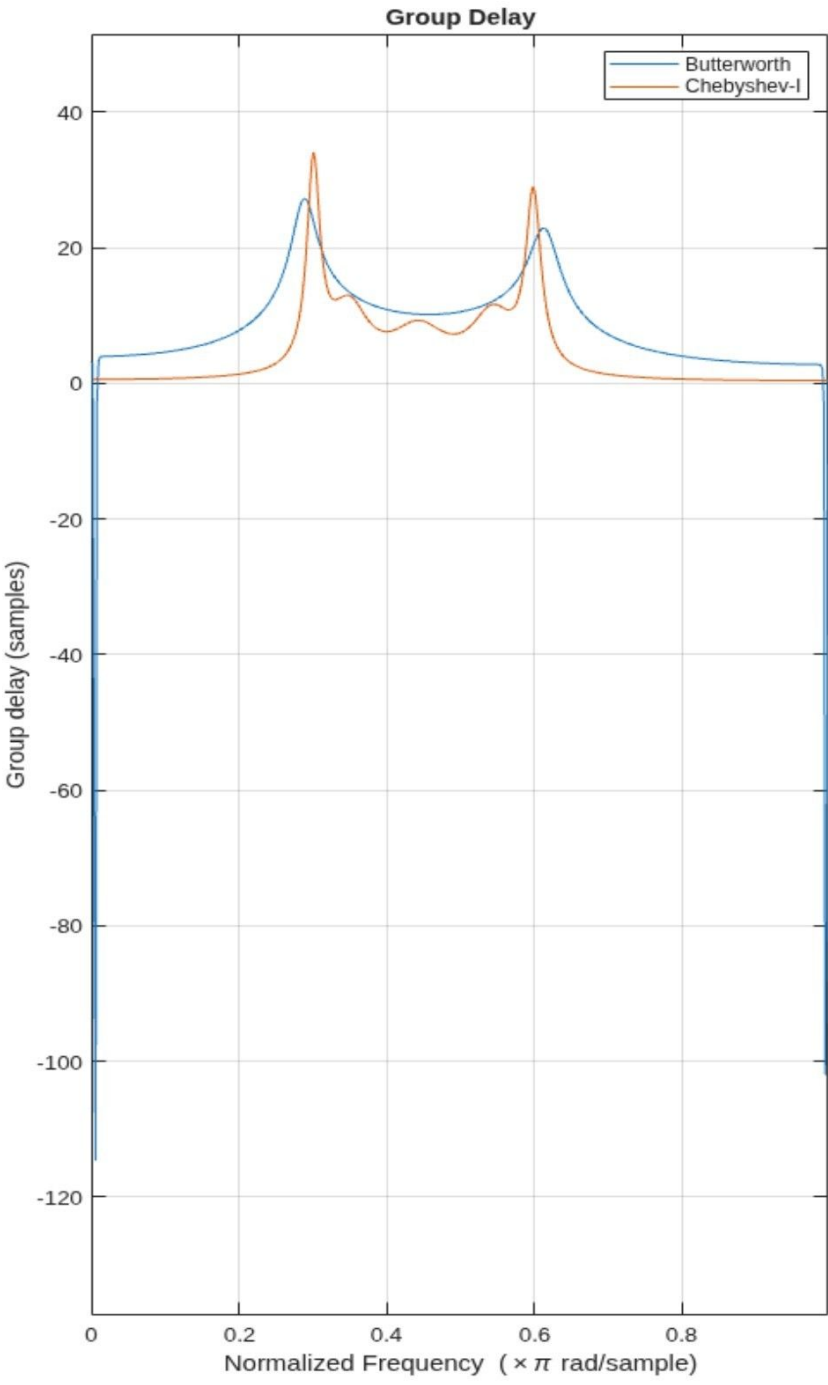


Figure 4: Dsp3_1

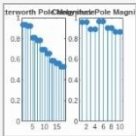
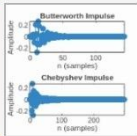
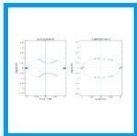
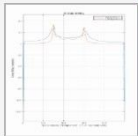
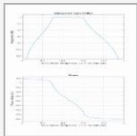
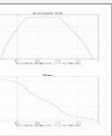
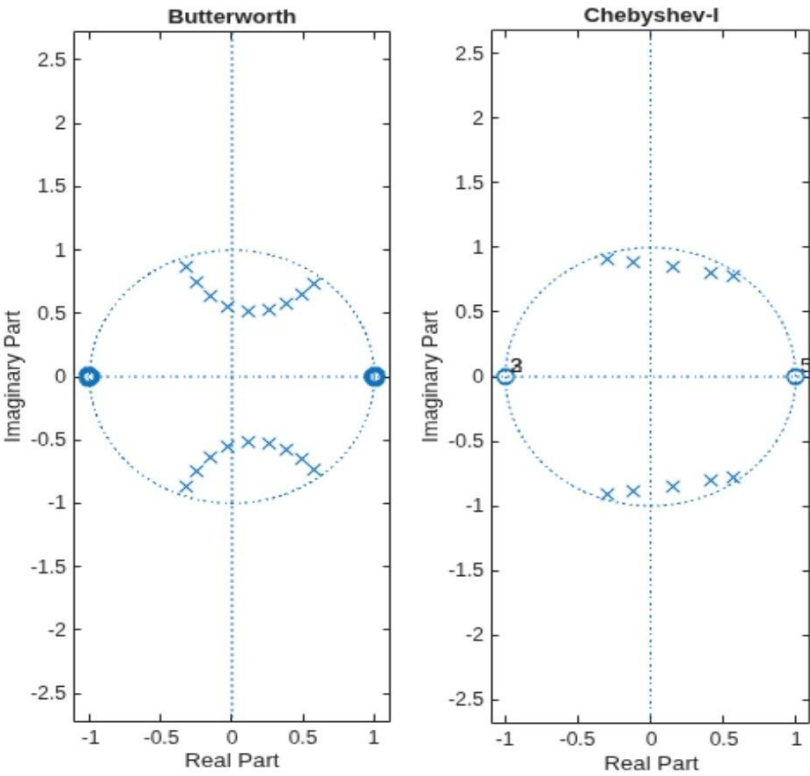


Figure 5: Dsp3_1

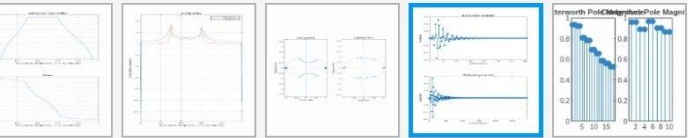
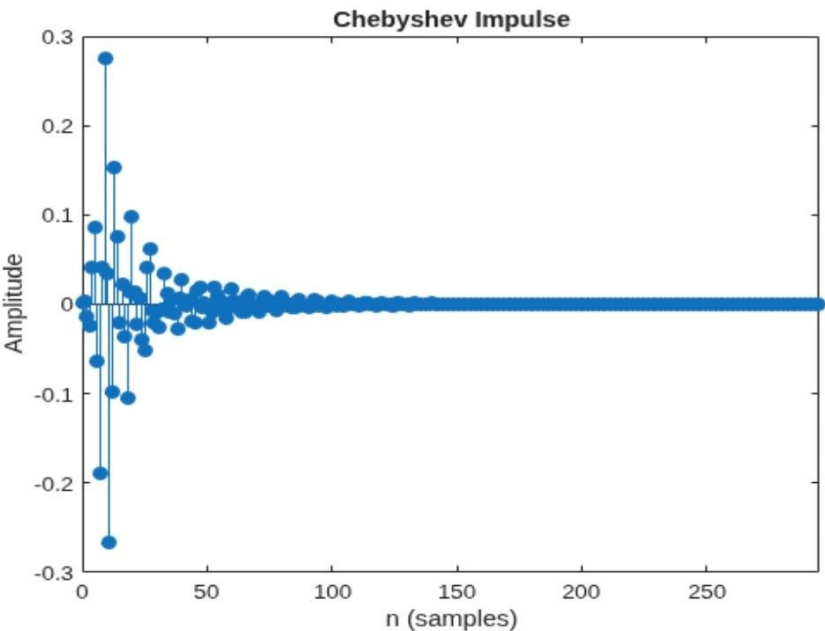
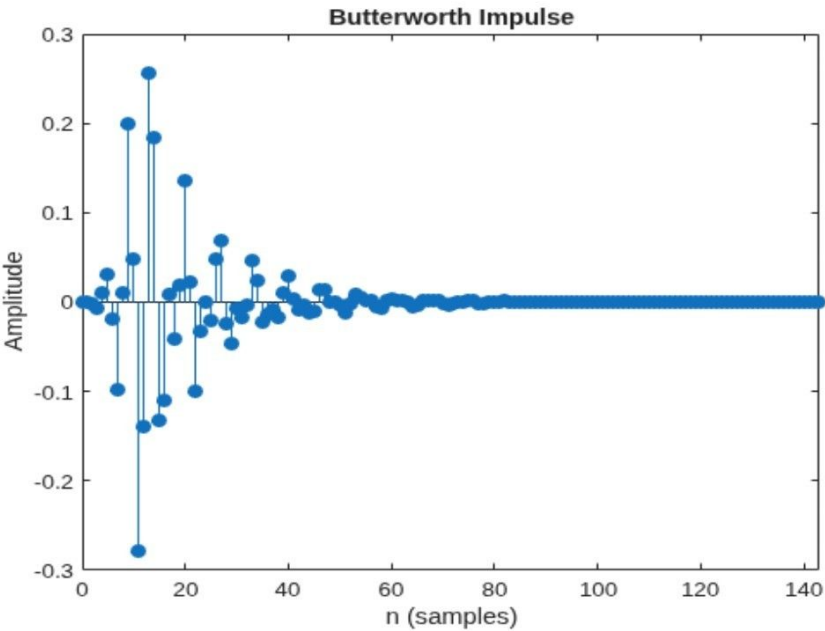


Figure 6: Dsp3_1

